

tests/ constitution_inheritance_gate.sh — operator guide

Revision: 1 **Last modified:** 2026-07-01T00:00:00Z **Authority:** Helix Constitution §11.4.35 (canonical-root inheritance clarity), §11.4.6 (no guessing — anchors derived from real submodule content), §1.1 (paired mutation), CLAUDE.md Hard Stop #1 (no CI/CD, no git hooks)

Companion (§11.4.18) to the in-source documentation block at the top of the script.

Overview / Purpose

The **pre-build / pre-merge inheritance gate**: it mechanically proves the Helix Constitution submodule is present and that this project genuinely inherits from it (§11.4.35), so a missing / broken / un-wired constitution submodule is caught BEFORE any build, merge, or release tag. It is read-only and self-contained, asserting five inheritance invariants via fixed-string (grep -F) checks against real submodule + parent files (the anchors are the SAME literals the shipped constitution/meta_test_inheritance.sh mutates, never guessed).

Usage

```
bash tests/constitution_inheritance_gate.sh
```

No CLI flags or arguments. Paths are derived from the script's own location.

Inputs

None (no env vars, no arguments). Files it reads:

- constitution/ directory (I1).
- constitution/Constitution.md — must carry ### §11.4 End-user quality guarantee — forensic anchor (I2).

- constitution/CLAUDE.md — must carry ## MANDATORY ANTI-BLUFF COVENANT (I3).
- constitution/AGENTS.md — must carry ### Anti-bluff covenant (I4).
- CLAUDE.md — must carry INHERITED FROM constitution/CLAUDE.md (I5a).
- AGENTS.md — must reference constitution/AGENTS.md (I5b).
- CONSTITUTION.md — must reference constitution/Constitution.md (I5c).

Outputs

- A header, one ✓ PASS/x FAIL line per invariant, and a === summary: N pass, M fail === line (fail details echoed to stderr).
- Exit 0 — all invariants hold; exit 1 — one or more FAILED.
- Colour is emitted only when stdout is a TTY.

Side-effects

None. Read-only; never mutates the tree.

Dependencies

- bash (set -uo pipefail), grep.

Related scripts

- tests/pre_build_verification.sh — the pre-build runner that invokes this gate.
- tests/test_constitution_inheritance.sh — the comprehensive host test that delegates to this gate (Part A) and adds recursive nested-submodule pointer checks (Part B).
- constitution/meta_test_inheritance.sh — the generic §11.4-anchor mutation.
- challenges/scripts/meta_test_constitution_inheritance.sh — the paired §1.1 mutation proving THIS gate catches regressions.
- tests/run-tests.sh — runs this gate first via test_constitution_inheritance().

Last verified

2026-07-01 — documented from source; the five invariants + their exact literal anchors read directly from the script body. Not executed here.